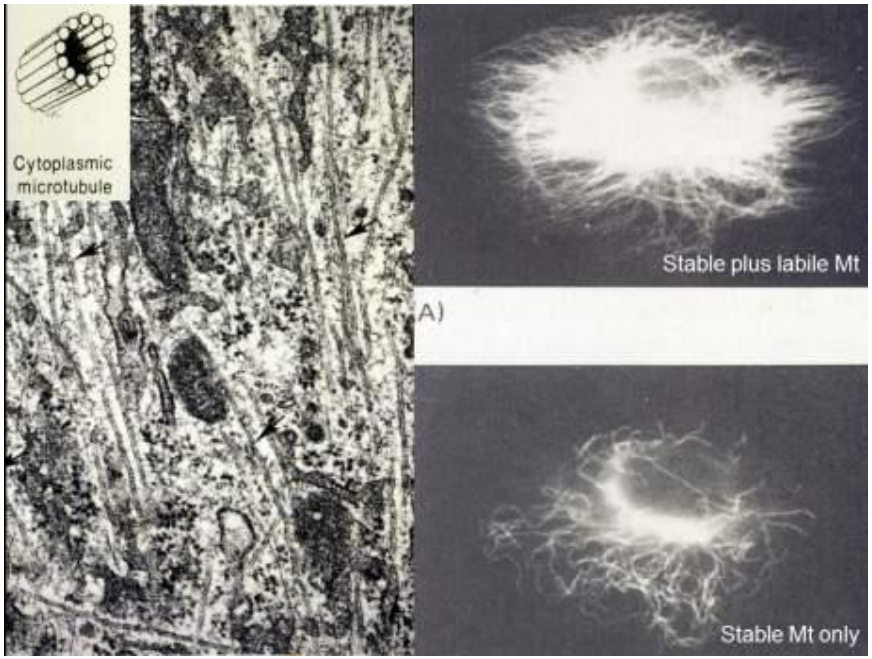
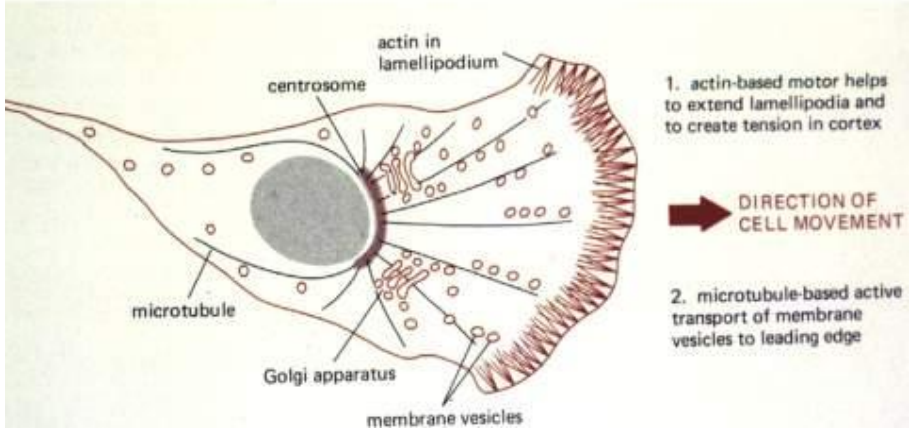
Conference on the CYTOSKELETON



MICROFILAMENTS - FUNCTION

- CELL MOTILITY - ACTIN AND MYOSIN



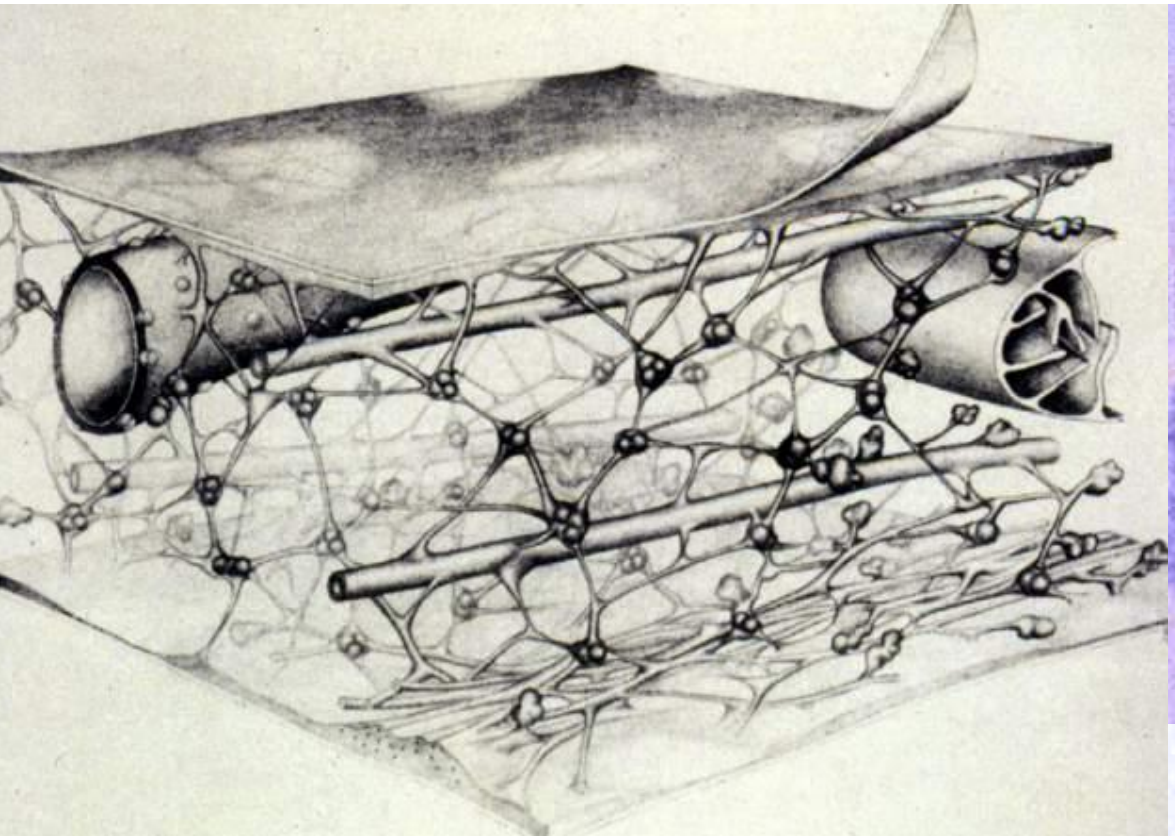
CYTOSKELETON

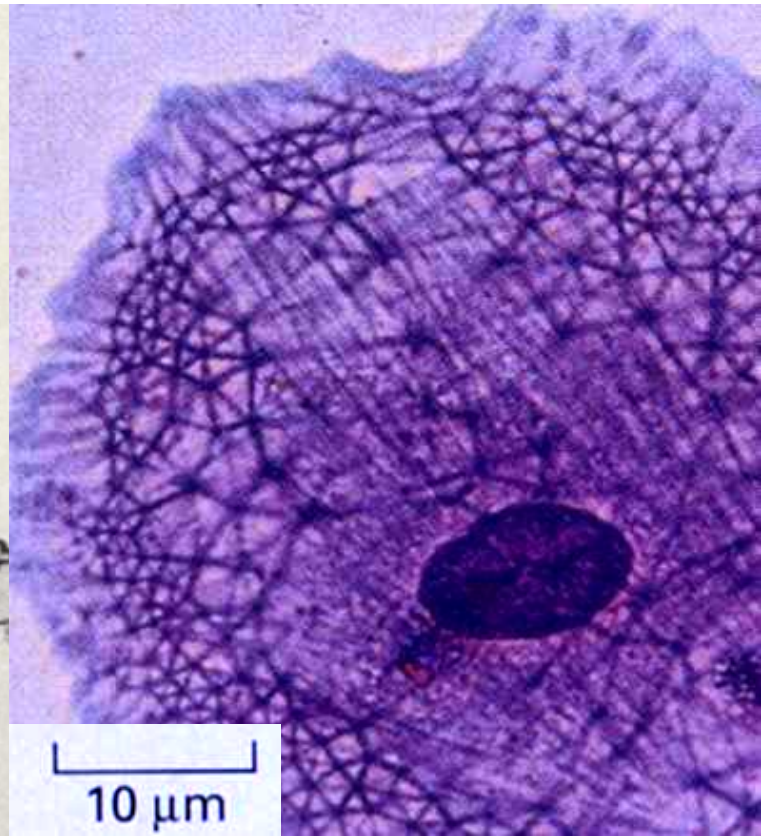
NON-MEMBRANOUS ORGANELLES

MICROTUBULES (25 nM)

MICROFILAMENT (6 nM)

INTERMEDIATE FILAMENT (10 nM)





Objectives of the Conference on the Cytoskeleton

- To reinforce the idea that the cytoskeleton provides structural framework and contractile machinery
- To reinforce the idea that MAPs provide stability to microtubules while providing support and enzymatic activity.
- To use information on drugs that break down different cytoskeletal components to solve problems and predict outcomes of drugs on cellular responses
- To reinforce the idea that defects in cytoskeletal components can provide opportunities for diseases

review

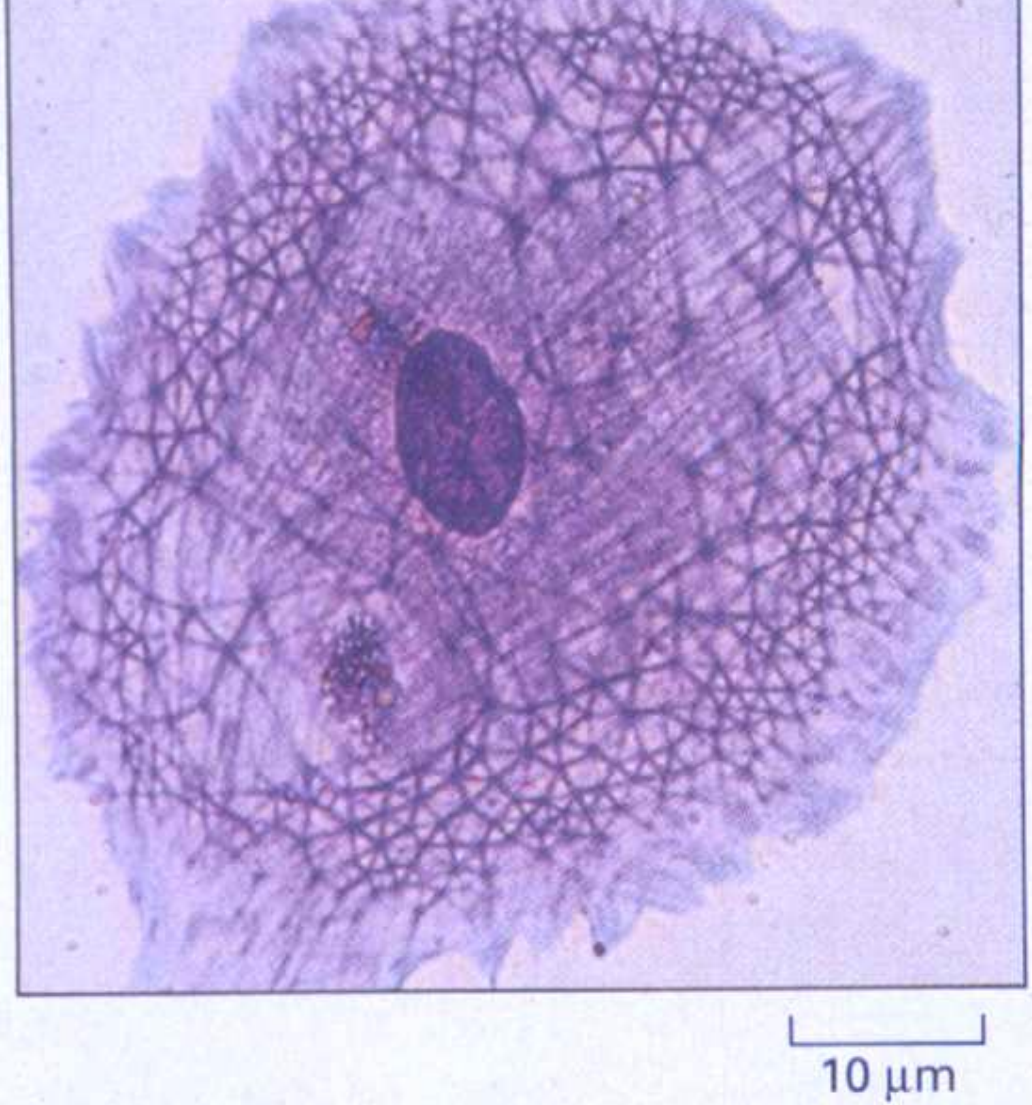
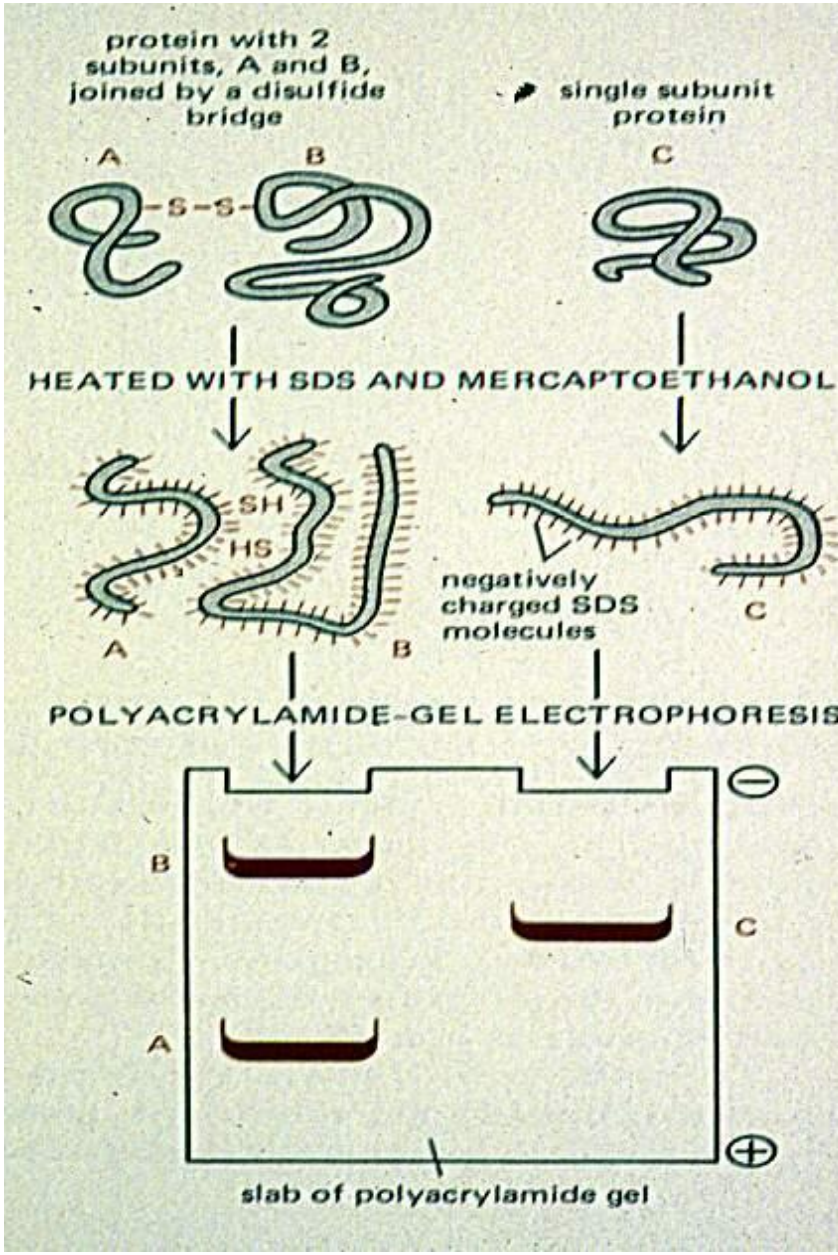


Figure 16–1 The cytoskeleton. A cell in culture has been fixed and stained with Coomassie blue, a general stain for proteins. Note the variety of



CYTOSKELETON –

Cells must perform tasks: REQUIRING

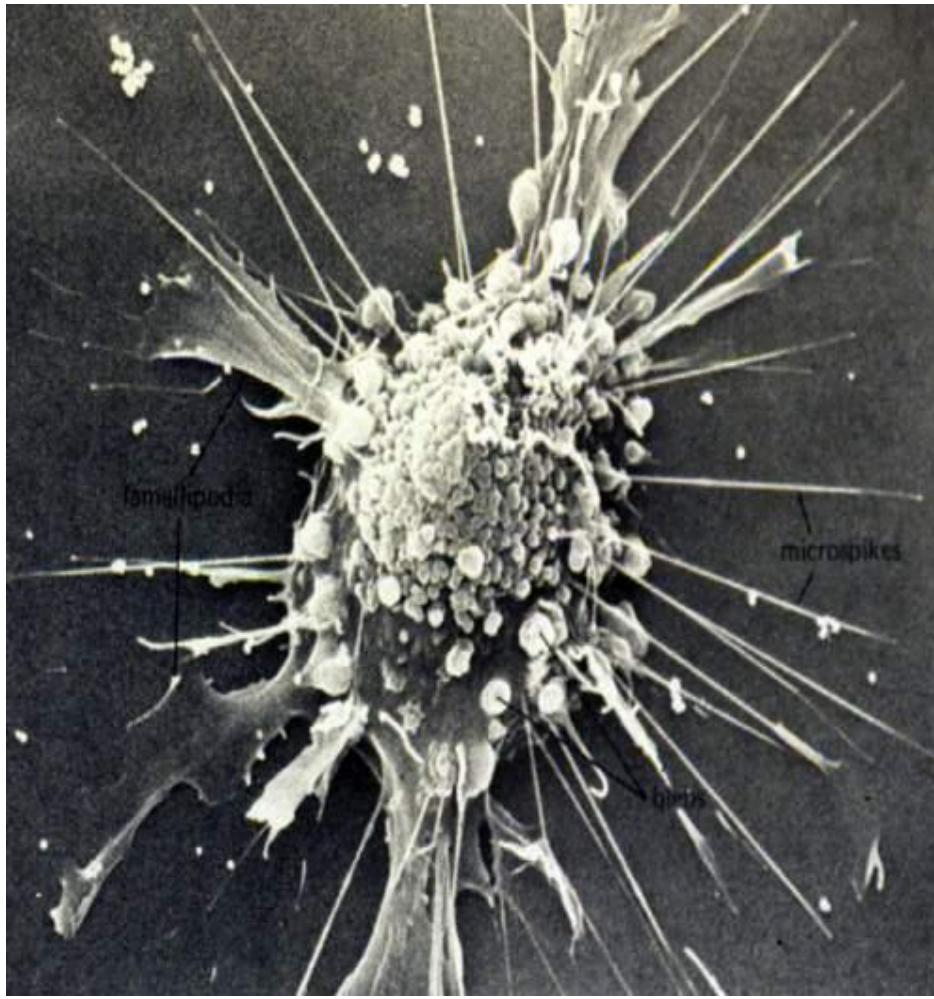
STRUCTURAL FRAMEWORK

CELL MAINTAIN SHAPE

CHANGE SHAPE

ENDOCYTOSIS AND PHAGOCYTOSIS

STABILIZATION OF CELL ATTACHMENT



CYTOSKELETON

Cells must perform tasks: REQUIRING

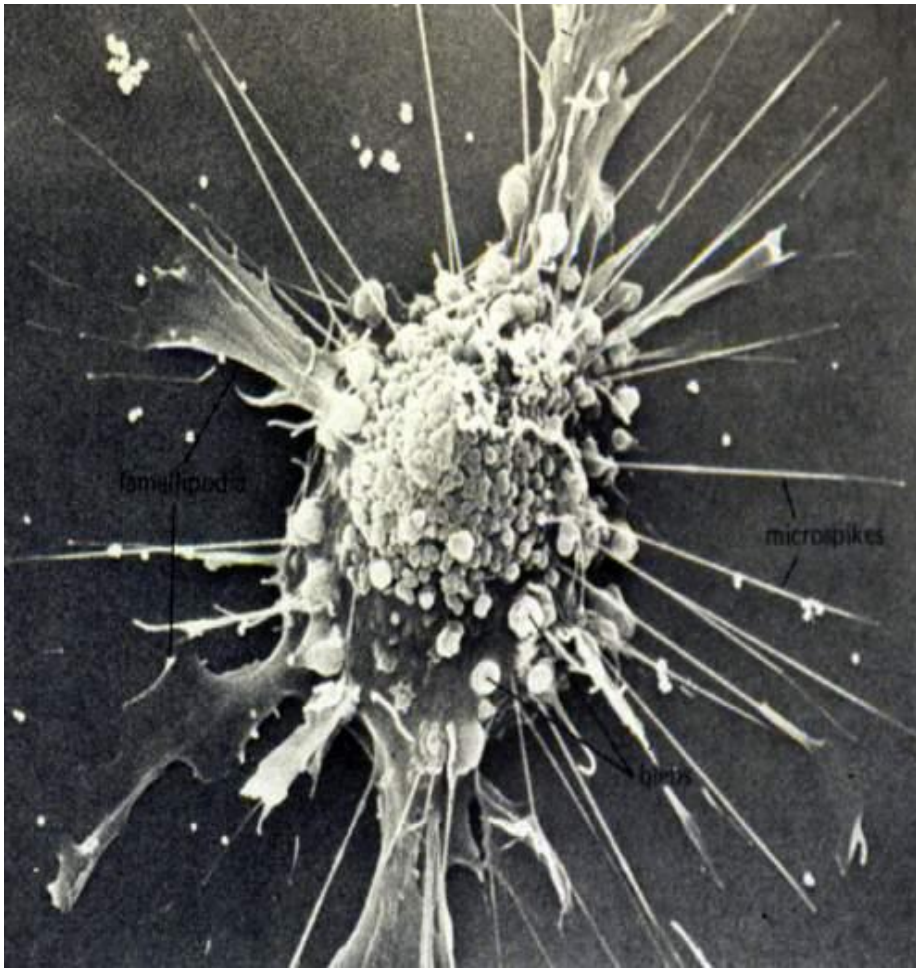
CONTRACTIAL MACHENARY

TRANSPORT VESICALS, ORGANELLES, AND CHROMOSOMES

DIVIDE CYTOPLASM

MOVEMENT OF LOCAL SPECIALIZATION OF CELL SURFACE

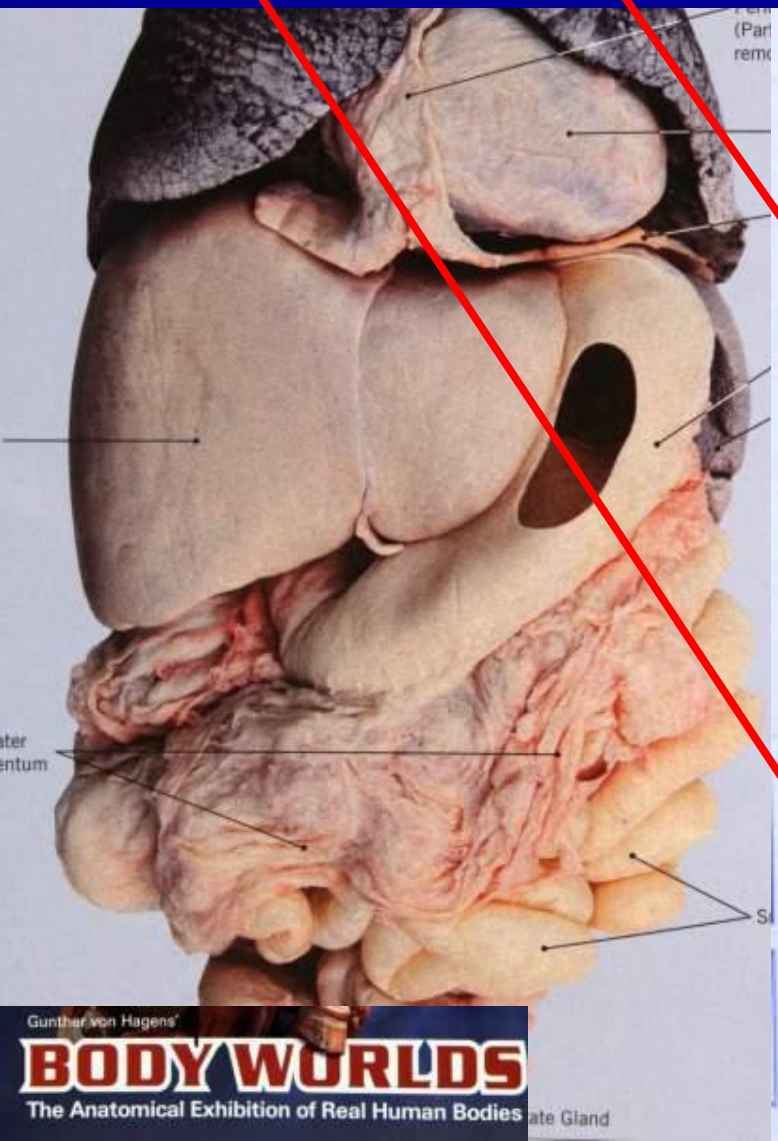
CELL MOTILITY

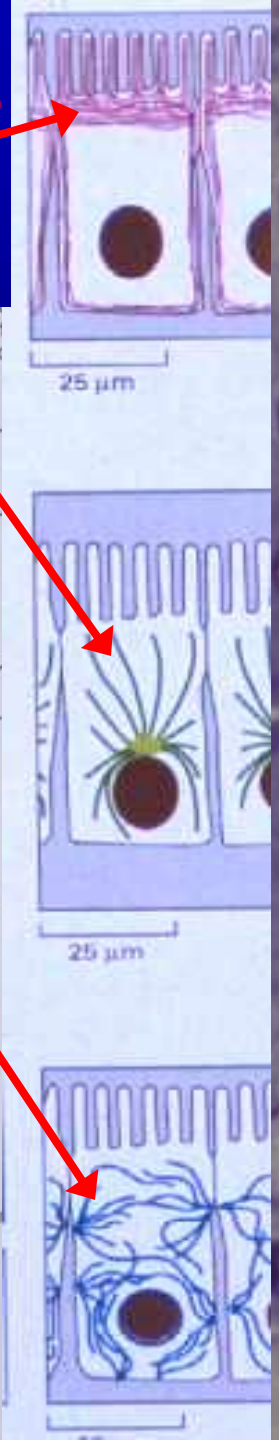


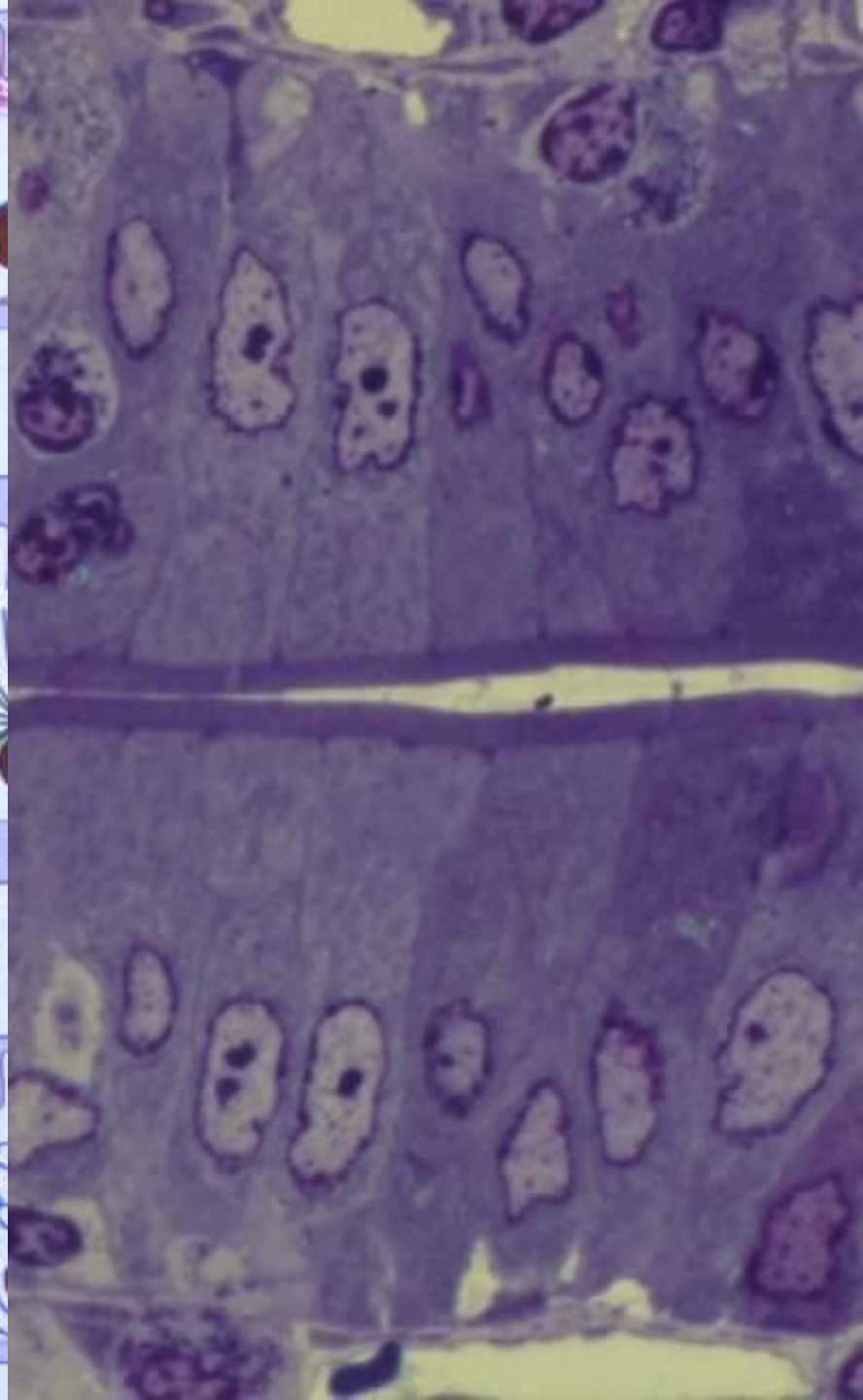
CYTOSKELETON

NON-MEMBRANOUS ORGANELLES

- MICROTUBULES (25 nM)
- MICROFILAMENT (6 nM)
- INTERMEDIATE FILAMENT (10 nM)

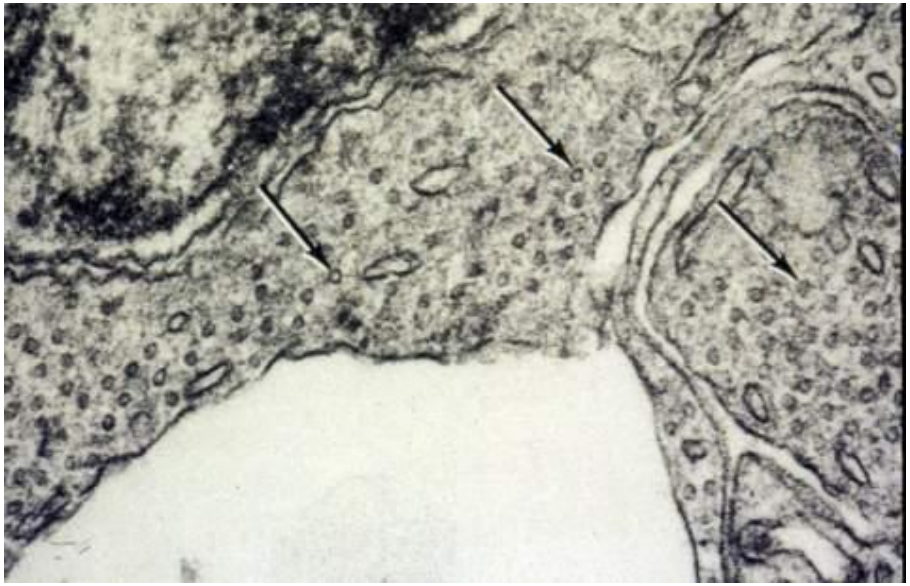


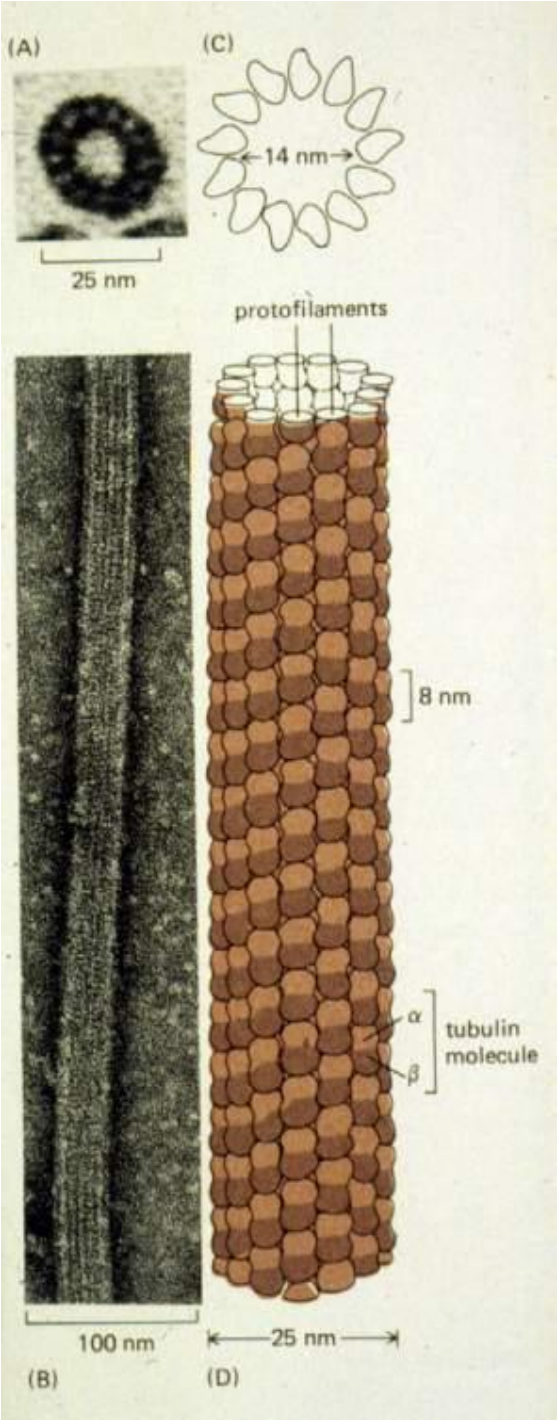


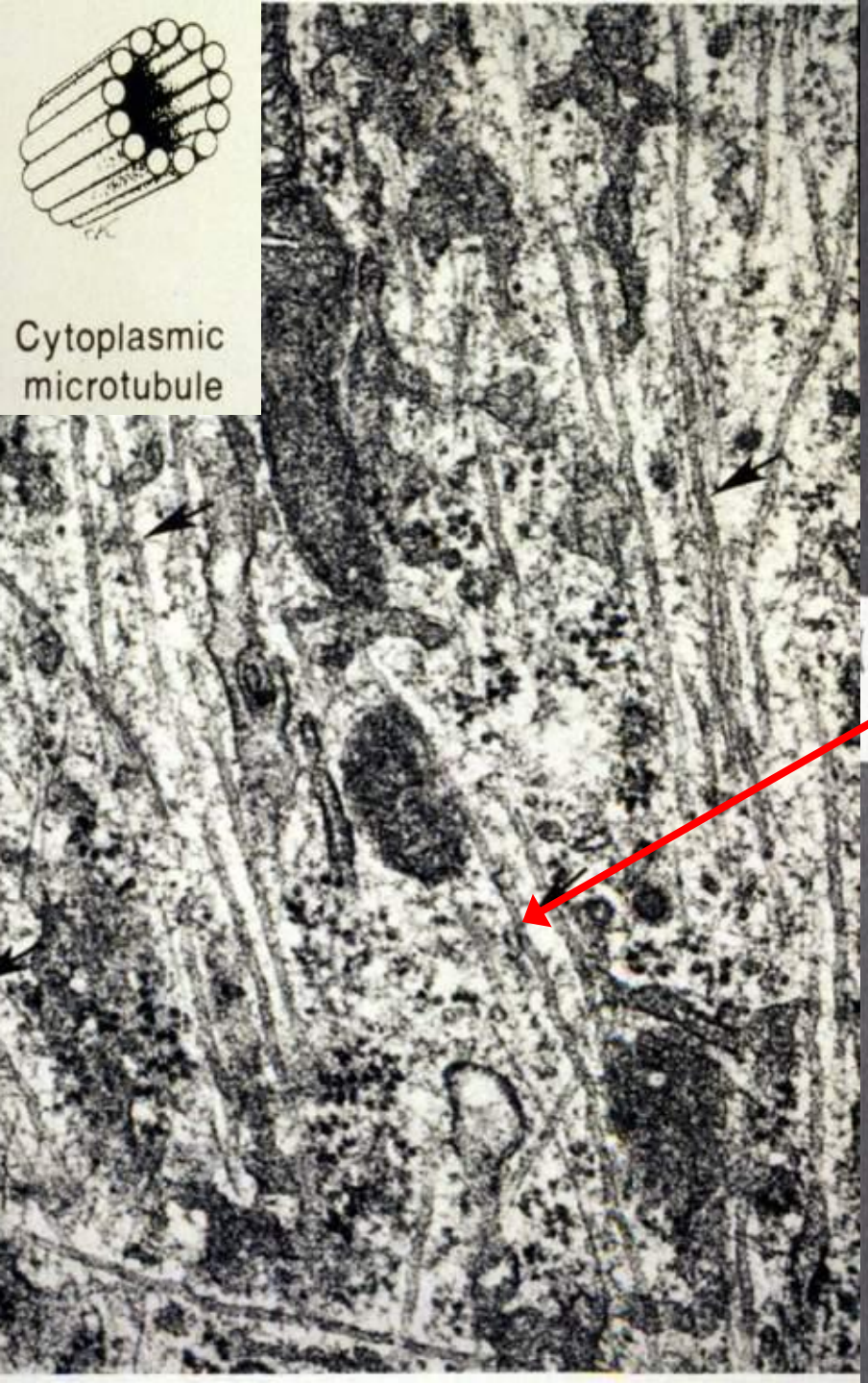


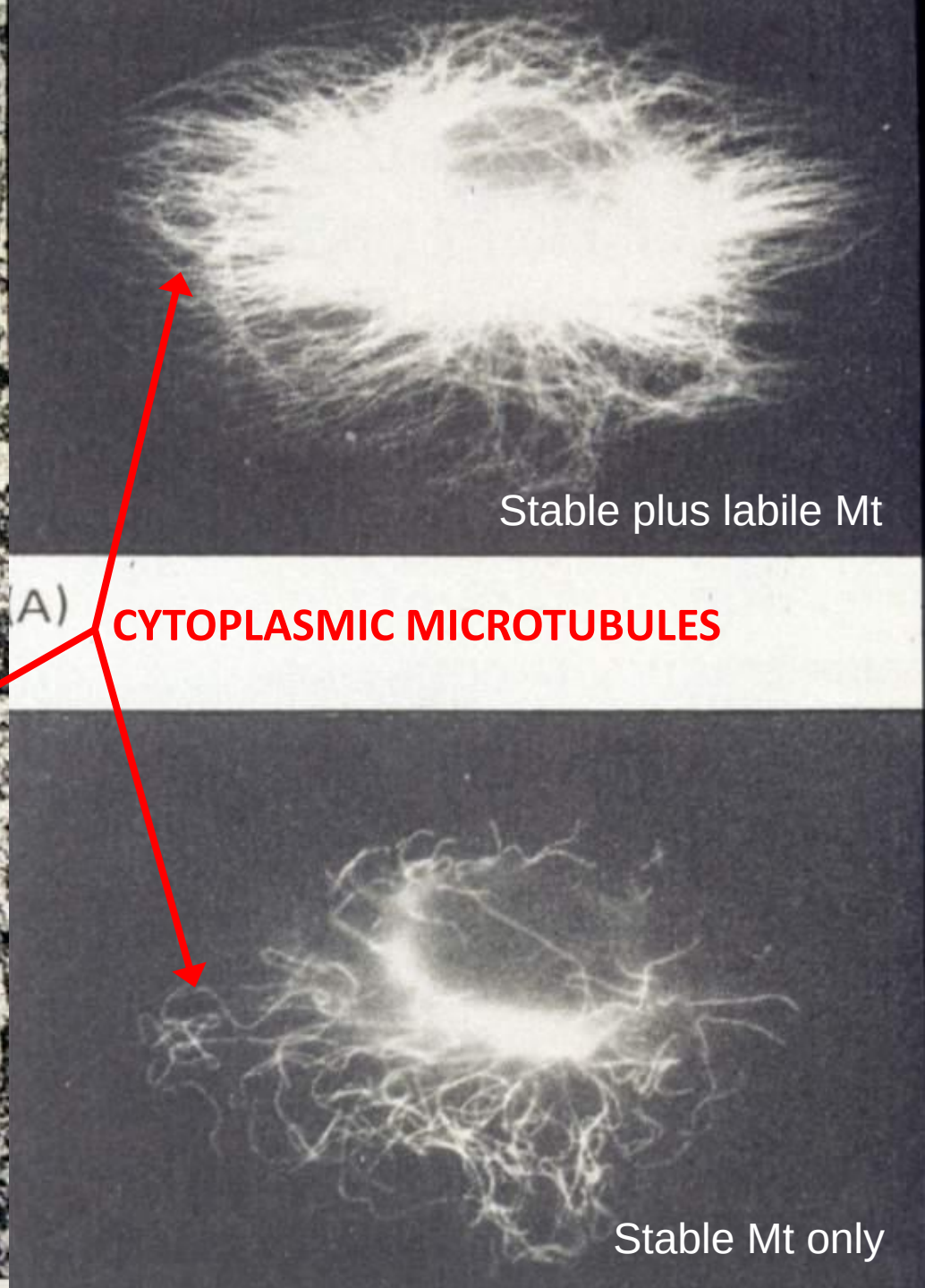
MICROTUBULE FUNCTION

- GUIDE CONTRACTILE FORCE (ACTIN) TO MOVE THE CELL ORGANELLES WITHIN ITS CYTOPLASM
- ORGANIZATION OF GOLGI, ER, AND MITOCHONDRIA
- SEPARATE CHROMOSOMES DURING MITOSIS

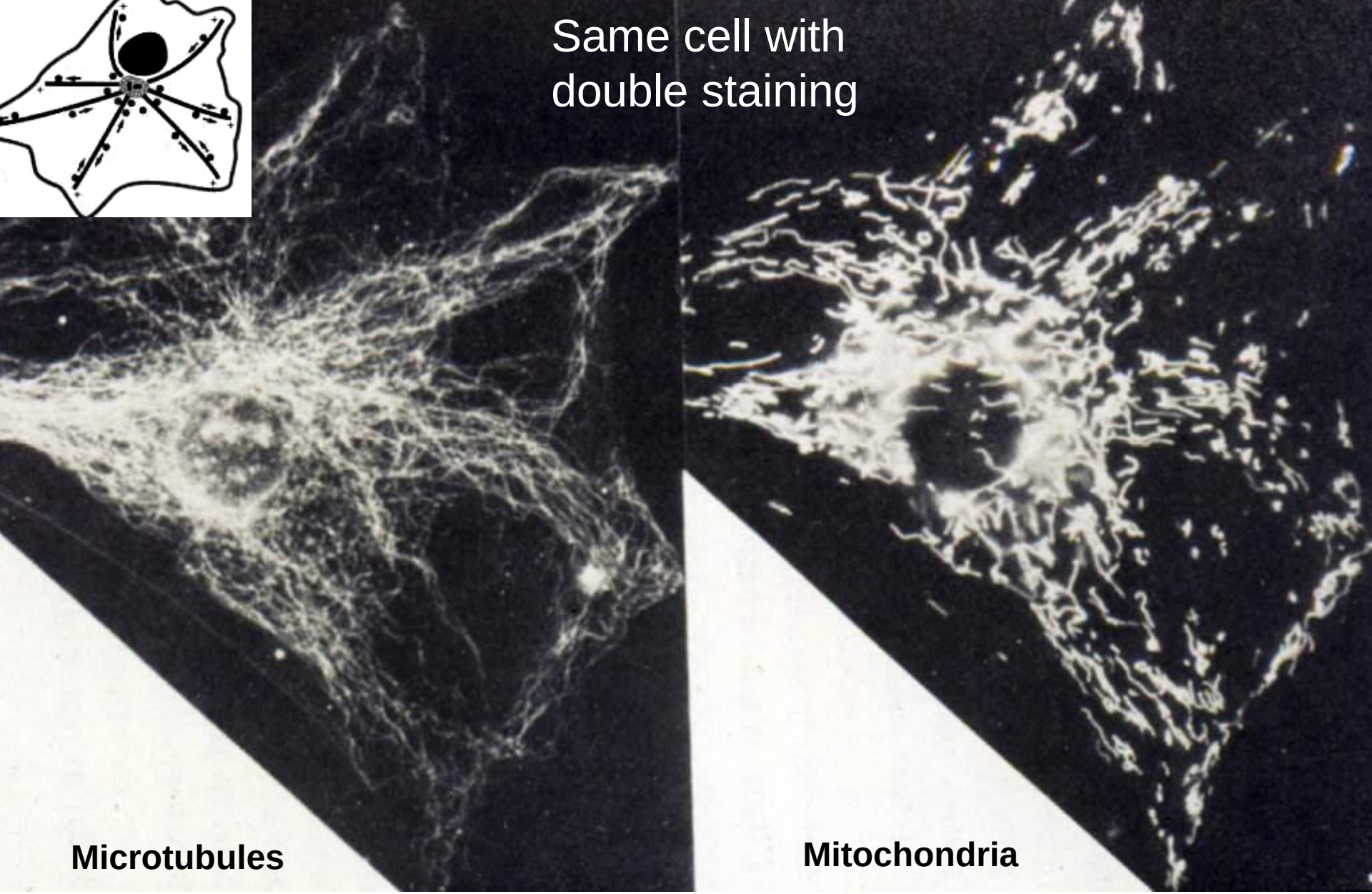








CYTOPLASMIC MICROTUBULES

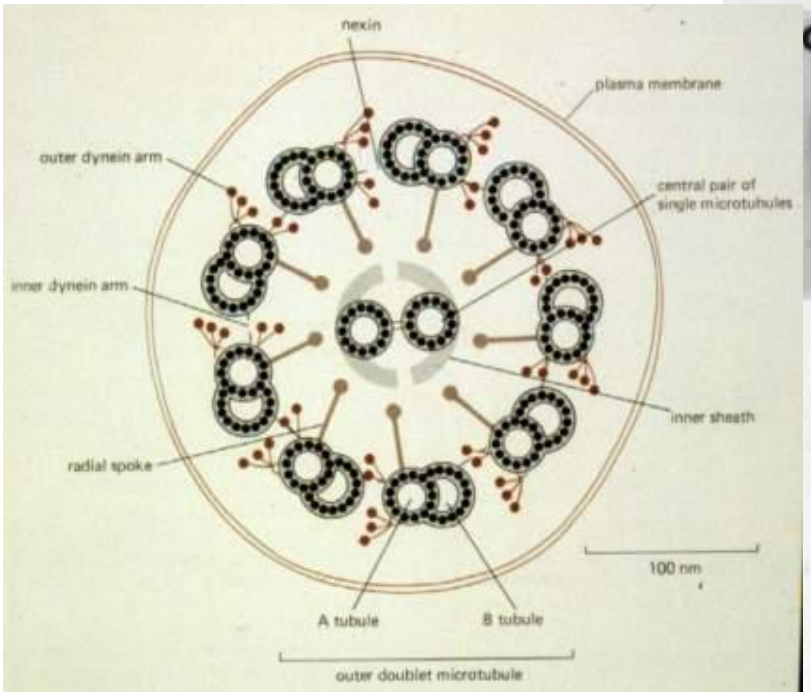

Same cell with
double staining
Microtubules
Mitochondria

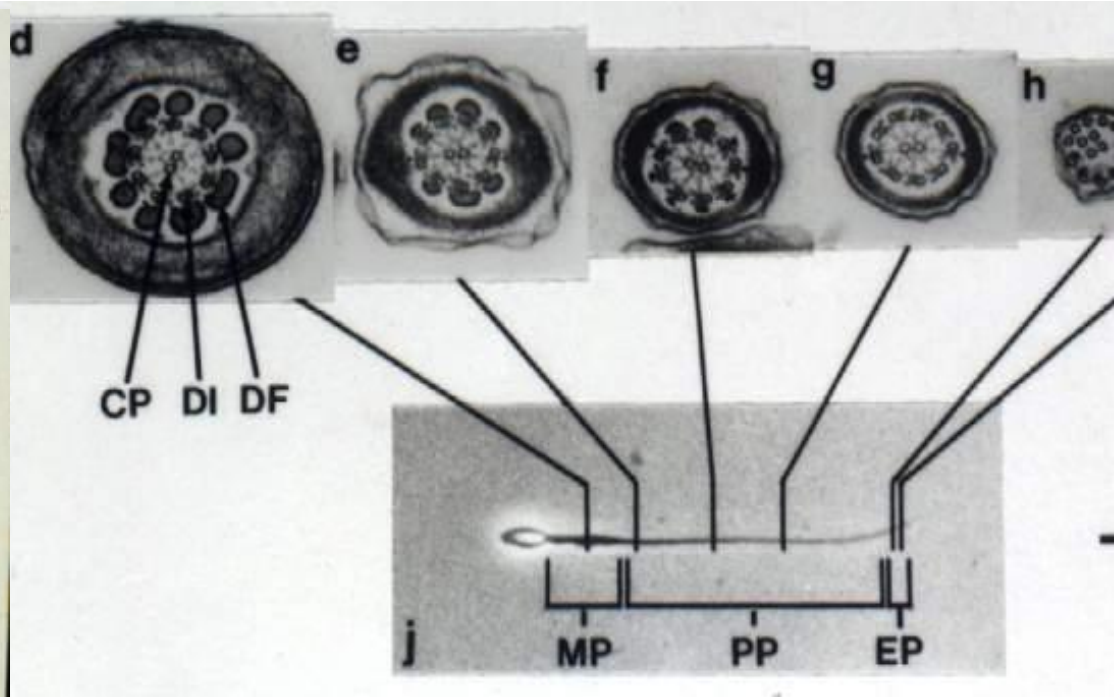
AXONEME OF CILIA AND FLAGELLA

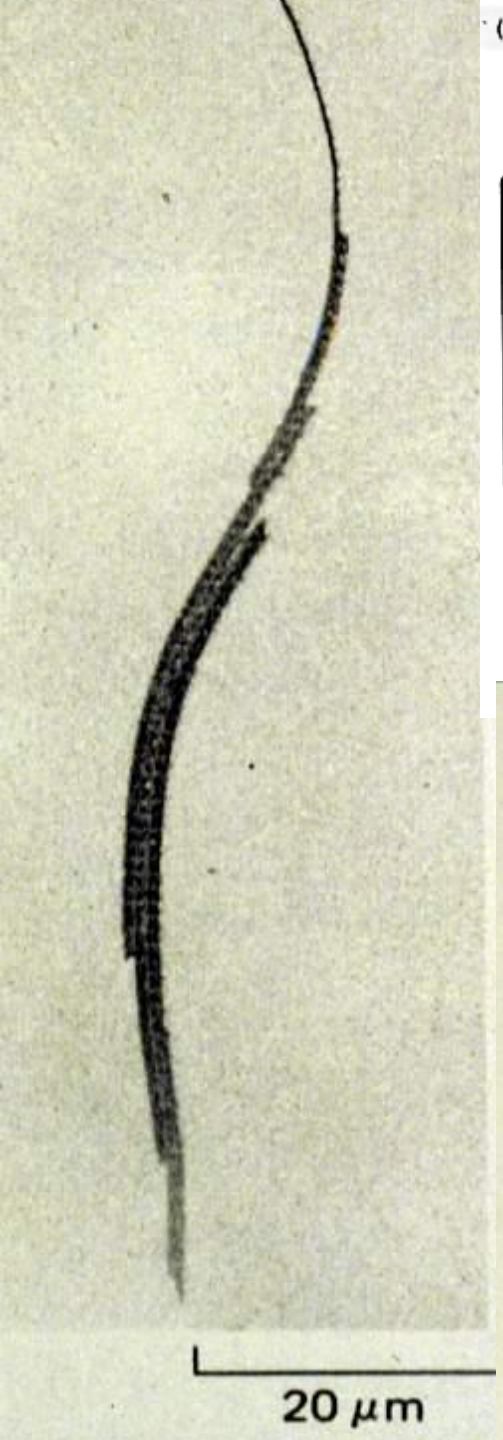
STABLE

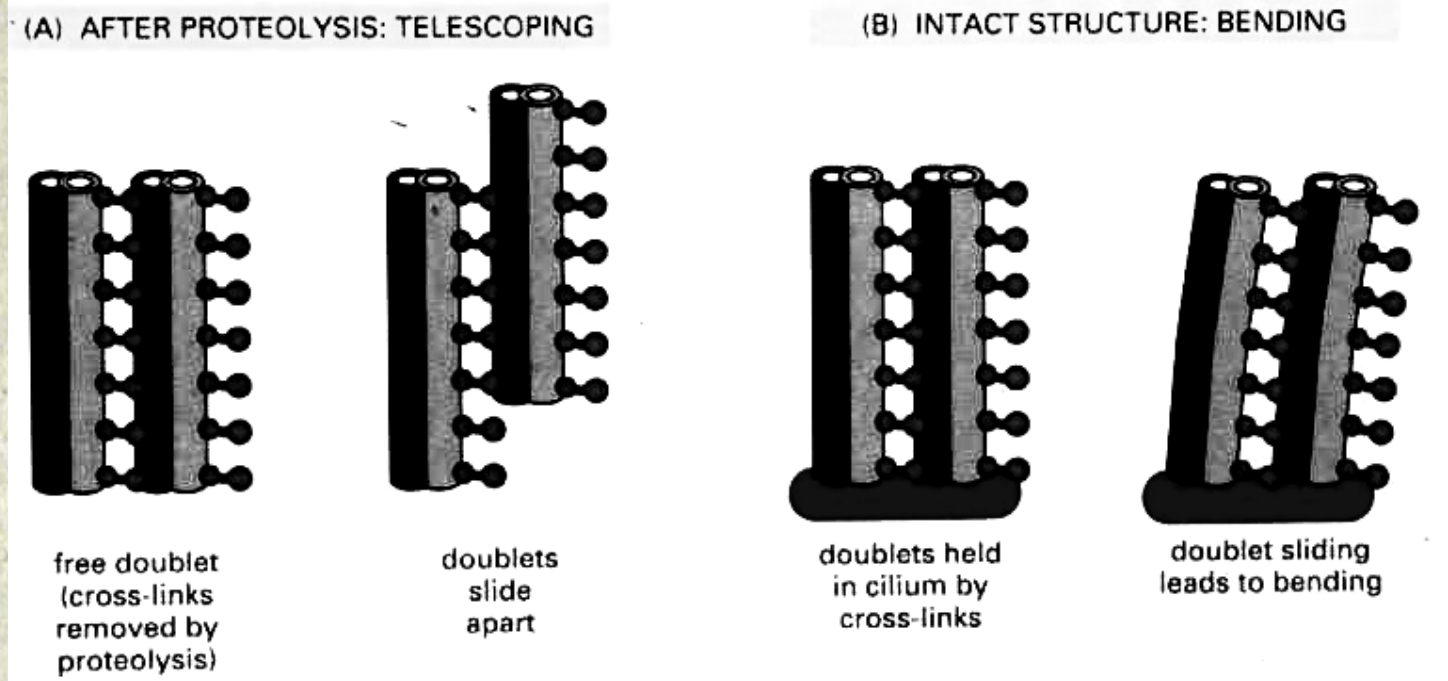
DYNEIN ARMS

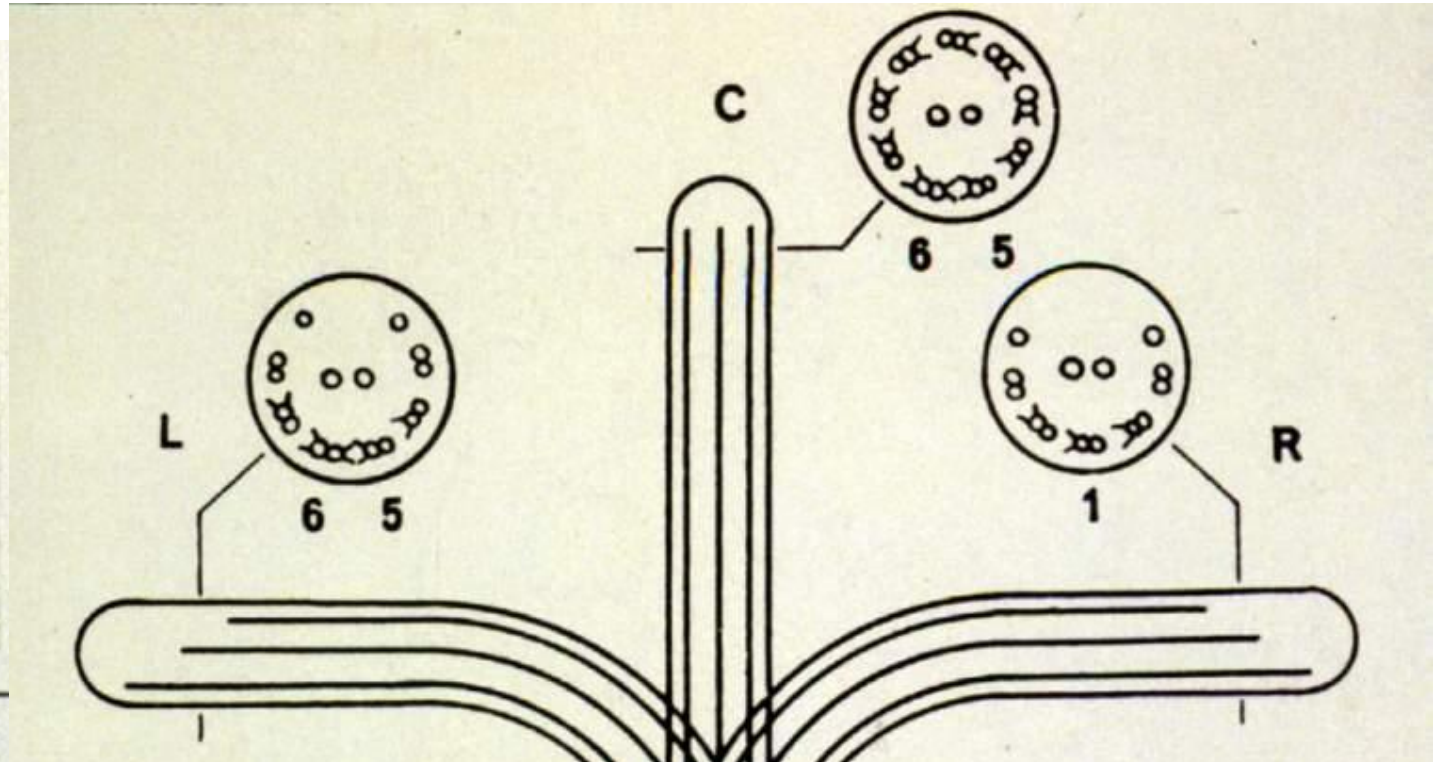
- **PAIRED LATERAL APPENDAGES**
- **PROTEIN ATPase ACTIVITY FOR CILIARY AND FLAGELLAR MOTILITY**






20 μm


(A) AFTER PROTEOLYSIS: TELESCOPING
(B) INTACT STRUCTURE: BENDING
free doublet (cross-links removed by proteolysis)
doublets slide apart
doublets held in cilium by cross-links
doublet sliding leads to bending


C
6 5
L
6 5
R
1

MICROTUBULES - SUMMARY

MICROTUBULES - CYLINDRICAL WALLED TUBES COMPOSED 13 PARALLEL PROTOFILAMENTS

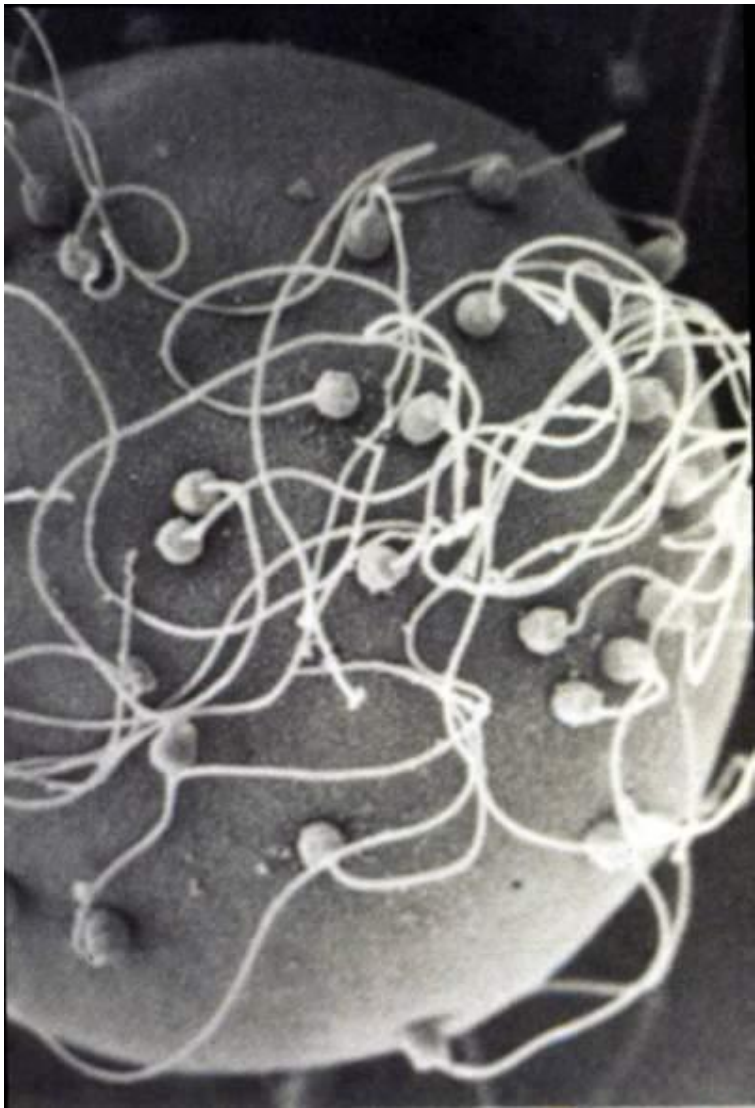
PROTOFILAMENTS - LINEAR POLYMERS OF ALPHA AND BETA TUBULIN

GROWTH AT ONE END AWAY FROM NUCLEATION SITE

POLARITY OF DIRECTION OF GROWTH - DIRECTS MOVEMENTS OF CYTOPLASMIC ORGANELLES

MICROTUBULE ASSOCIATED PROTEINS - FORM STABILIZING CROSS LINKS OF MT AND MAY BE ASSOCIATED WITH POLYMERIZATION OF MICROTUBULES

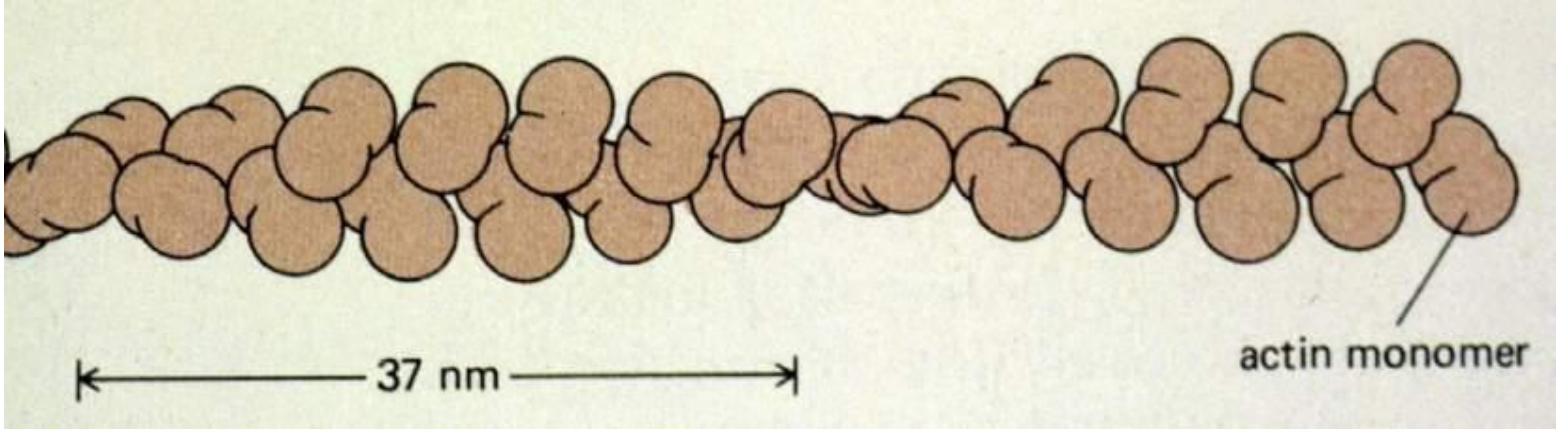
COLCHICINE



MICROFILAMENTS

COMPOSITION:

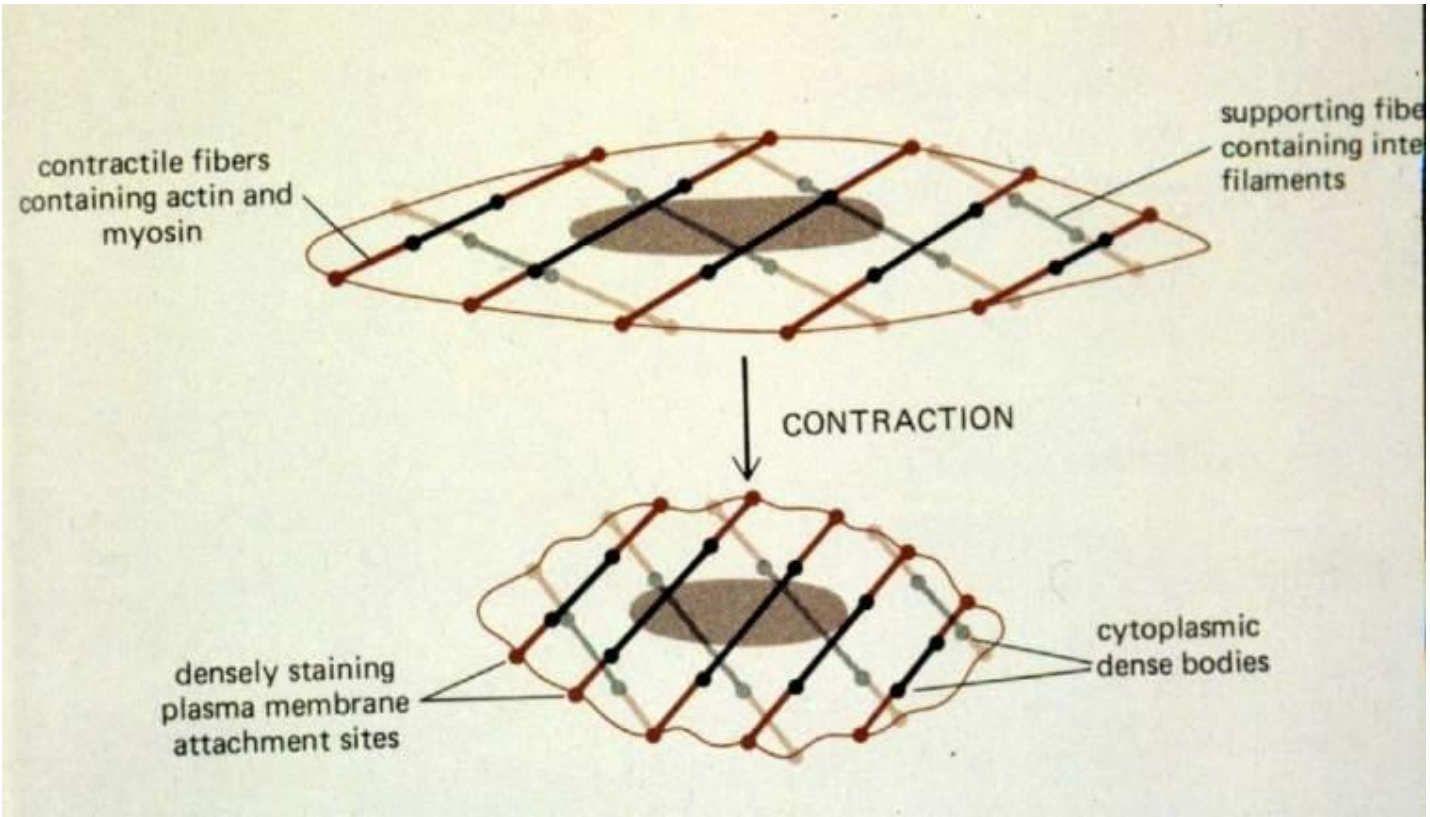
- **ACTIN - HIGHLY CONSERVED PROTEIN**
- **ACTIN ASSOCIATED PROTEINS**
  - **MYOSIN - ATPase**
  - **TROPHOMYOSIN - ROD-LIKE PROTEIN**
  - **FILAMIN - BUNDLES ACTIN FILAMENTS**



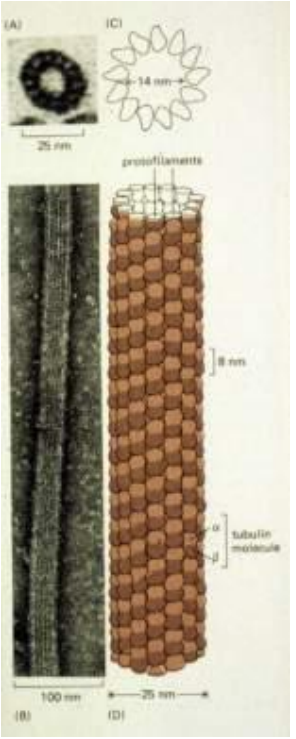
MICROFILAMENTS - CONTRACTILE PROTEINS

ACTIN AND MYOSIN – MUSCLE AND MOST ALL CELLS

ACTIN 10% TO 15% OF CELLULAR PROTEIN, WIDELY DISTRIBUTED



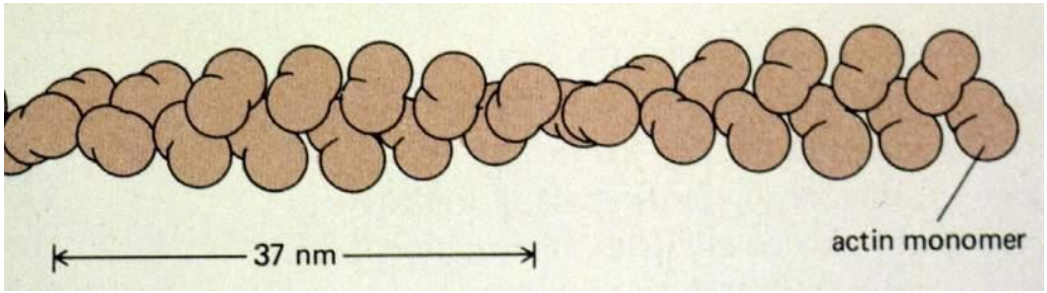
DRUGS THAT INFLUENCE - MICROTUBULES and microfilaments



INHIBITORS:

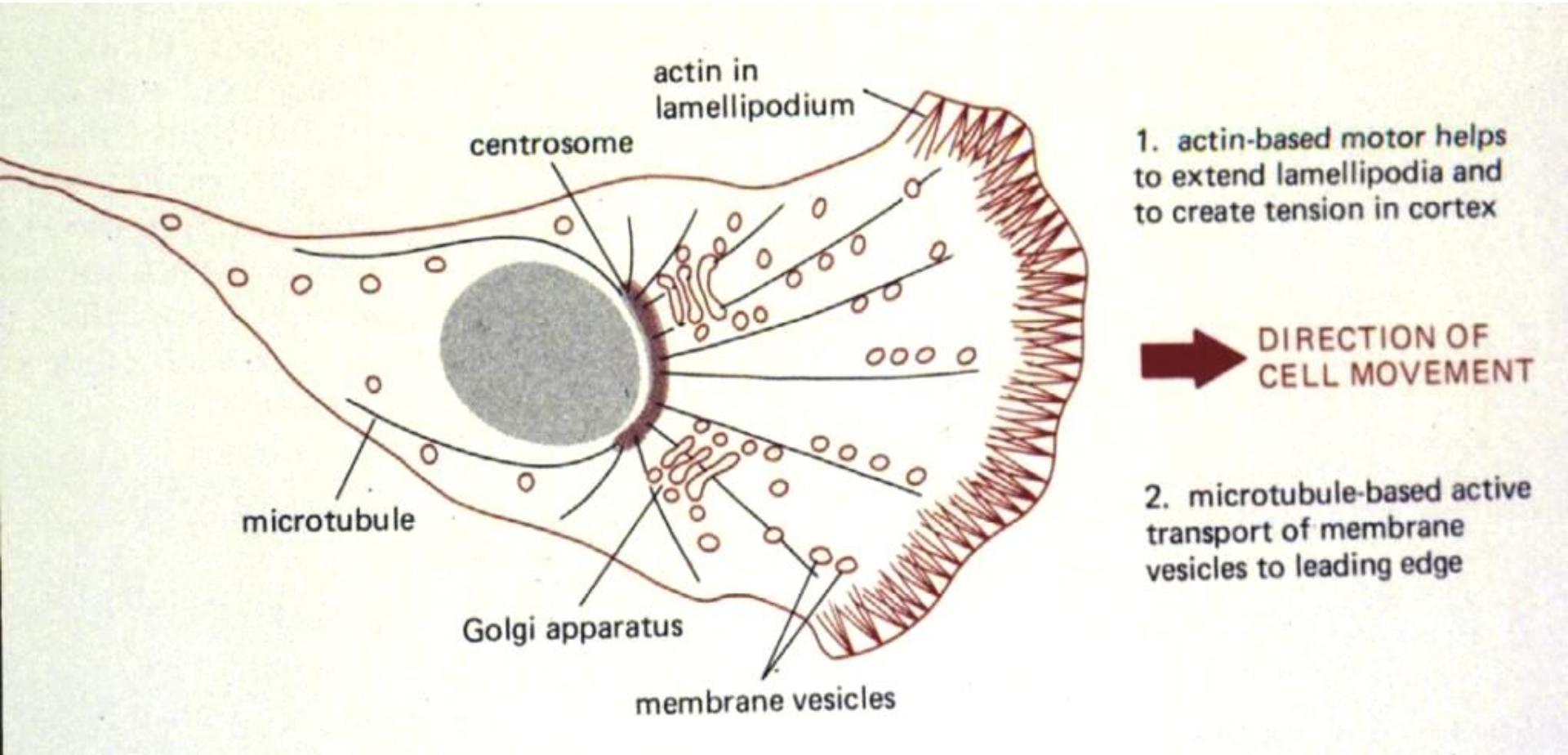
- **COLCHICINE** - INHIBIT ASSEMBLY <u>IN</u> <u>VITRO</u>, DESTROY <u>IN</u> <u>VIVO</u> MICROTUBULES

- **Cytochalasin B** - INHIBIT of microfilaments

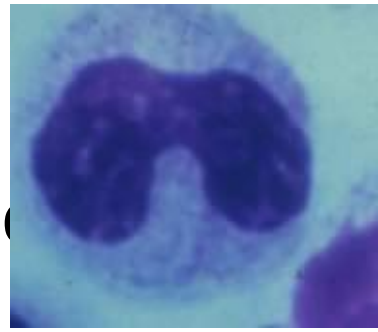


MICROFILAMENTS - FUNCTION

- **CELL MOTILITY - ACTIN AND MYOSIN**



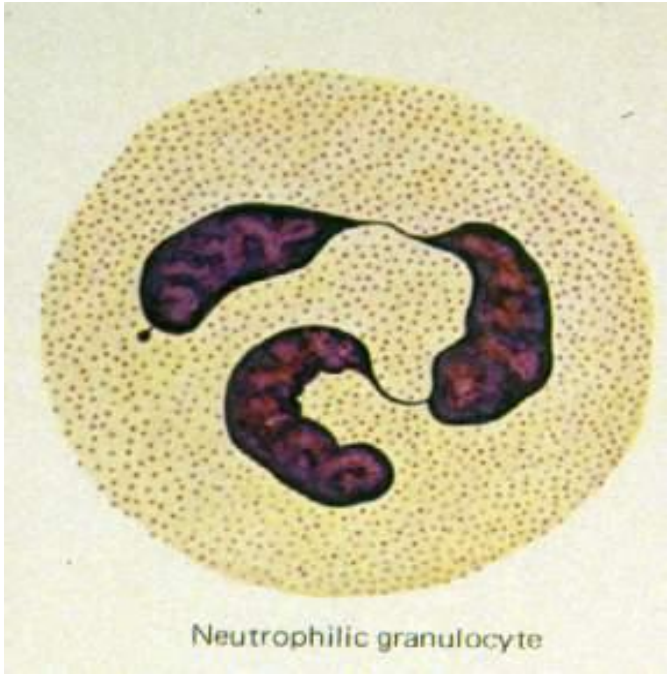
NEUROPHILS (PMN)



BAND CELLS - IMMATURE RELEASE OF CELLS INDICATIVE O
INFECTION

PHAGOCYTOSIS

- OPOSIN ENHANCED
- NONSPECIFIC



BASIC CELLULAR FUNCTIONS –

RANDOM MOTILITY,

CHEMOTAXIS,

PHAGOCYTOSIS,

KILLING BACTERIA

Considerations

- Structure of the axoneme and how it might look if defected
- Dynein has both structural and enzymatic activity, and a cell may be missing one or both
- Cilia on early embryologic cells may direct location of organs
- MAPs make microtubules stable, but labile microtubules direct vesicles
- Drug treatment can be used to evaluate the cytoskeletal component responsible for cell responses

CONFERENCE ON CYTOSKELETON

1. A male patient with a genetic disorder called Kartagener's syndrome was diagnosed with infertility. Although he had a normal sperm count, the sperm were immotile. When the sperm tails (flagella) were examined by electron microscopy, it was found that the axonemes lacked the structures which contain dynein.

   A. How would these defective flagella look when examined in the electron microscope?

   B. What experiments could you conduct to test if dynein was actually missing from these cells, or if it was merely not assembled?

   C. What other health problems might patients with this syndrome have?

   D. Normal sperm flagella can be caused to beat in vitro by permeabilizing the flagella with a nonionic detergent, and then adding exogenous ATP; the ATP is hydrolyzed as the microtubule doublets slide past each other. When sperm from the patient described above were tested in this assay, they failed to hydrolyze ATP. Sperm from another patient, diagnosed with Kartagener's syndrome, were also immotile and lacked dynein arms but were able to hydrolyze ATP. The flagella however did not beat. How could you explain the difference in the defect in each of the two cases?

   E. 50% of Kartagener's patients have their hearts on the right side instead of the left. Can you speculate on how an absence of dynein could lead to this condition.

2. A. In the presence of colchicine, cytoplasmic microtubules become depolymerized while microtubules in cilia do not depolymerize. How might you explain this phenomenon? What would you predict would happen to cytoplasmic and ciliary microtubules at 4°C?

   B. PMN leukocytes undergo directed migration (chemotaxis) toward certain chemotactic peptides. When the cells are treated with colchicine, the cells still migrate, but in a random fashion. When the cells are treated with cytochalasin B, they cease to migrate altogether. What does this tell you about the function of microtubules and microfilaments in chemotaxis?

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- To use information on drugs that break down different cytoskeletal components to solve problems and predict outcomes of drugs on cellular responses
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